A Hölder calculus of currents

Gaining regularity from antisymmetry

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Abstract

Distributions model a wide range of physical phenomena, but difficulties arise as soon as non-linear phenomena come into play — starting with the product of two distributions, which has in general no canonical meaning. For Hölder distributions on $\mathbb{R}^d$, the situation is sharp: a canonical product can be defined as soon as the regularities of the two factors compensate — that is, their sum is positive. A current generalizes such a distribution to a k-dimensional object in $\mathbb{R}^d$: a differential form whose coefficients are distributions, modeling surfaces or more singular geometric objects. For such objects geometry changes the picture: the antisymmetry of the wedge structure can generate cancellations, making certain products canonically meaningful even when the sum of the regularities is nonpositive. We then build a Hölder calculus of currents, where these cancellations and the Reconstruction Theorem combine to gain regularity, and apply it to the Jacobian currents associated to Hölder functions, recovering the classical Young and Züst integrals.